

M.Tech, CSE (AI and ML)

Semester Wise Course Structure & Detailed Syllabus (2024-2025)

1ST SEM							
THEORY							
SL. NO.	PAPER CODE	PAPER NAME	L	T	P	CONTACT HRS./WEEK	CREDIT
01	PGCSPC101	Mathematical Foundation of Computer Science	3	0	0	3	3
02	PGCSPC102	Advanced Data Structure	3	0	0	3	3
03	PGCSPC103	Machine Learning	3	0	0	3	3
04	PGCSPC104	Artificial Intelligence	3	0	0	3	3
05	PGCSHS101	Universal Human Values	2	0	0	2	2
PRACTICAL							
01	PGCSPCL105	Advanced Data Structure Lab	0	0	3	3	1.5
02	PGCSPCL106	Machine Learning Lab	0	0	3	3	1.5
		Total	14	0	6	20	17

L – Lecture (total lecture 36), T – tutorial, P – practical

Name of the Course		Mathematical Foundation of Computer Science	
Course Code: PGCSPC101		Semester: 1 st	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: 3 hrs./week		Mid Semester 1 Exam: 15 Marks	
Tutorial Contact Hrs.: 0 hrs./week		Mid Semester 2 Exam: 15 Marks	
Practical: 0 hrs./week		Other Assessment tools (Assignment, Quiz etc.): 20 Marks	
Credit Points: 3		End Semester Exam:100 Marks (To be mapped into 50 mark)	
Objective:			
1.	To learn the concept of Eigenvalues, Eigenvectors diagonalization of matrices for understanding Engineering problems.		
2.	To understand linear spaces its basis and dimension with corresponding applications in the field of computer sciences		
3.	To understand Fourier series representation of Periodic signals.		
4.	To understand concept of random variable and probability distribution.		
5.	To understand counting techniques and combinatorics in the context of discrete probability.		
6.	To learn recurrence relations and generating functions.		
7.	To understand Algebraic structures and classify Boolean function.		
Pre-Requisite			
1.	Discrete Mathematics		
2.	Engineering Mathematics (UG level)		
Detailed Syllabus			
Module	Content	Hrs	Marks
1	Advanced Linear Algebra: Vector space, Definitions and examples, Subspace, Union and intersection of subspaces, Linear sum of two subspaces, Linear combination, independence and dependence, Linear span, Generators of vector space, Finite dimensional vector space, Replacement Theorem, Extension theorem, Statement of the result that any two bases of a finite dimensional vector space have same number of elements. Dimension of a vector space, Extraction of basis, formation of basis with special emphasis on R_n ($n \leq 3$), Eigen value and eigenvectors of matrices, Caley Hamilton Theorem, Simple properties of eigen values and eigen vectors- for symmetric and general matrices, Diagonalisation.	10	
2	Counting Techniques: Pigeon- hole Principle, Principles of inclusion and exclusions; Recurrence relations: Formulation & Modelling of different counting problems in terms of recurrence relations, Solution of linear recurrence relations with constant coefficients (up to second order) by (i) The iterative method (ii) Characteristic roots method (iii) Generating functions method.	4	
3	Stochastic Process: Random variable, Distribution function, Interval estimation, Confidence intervals, Test of hypothesis, Markov process with special emphasis on Markov chain.	6	
4	Integral Transform: Fourier Series and Transform: Periodic functions, Trigonometric functions, Trigonometric Series, Fourier series, Dirichlet conditions, Euler formula for Fourier coefficients, Even and Odd functions, Half range series expansion, Parseval's formula. Fourier transform, Properties of Fourier transform, Fourier sine and cosine transform, Convolution theorem, First Fourier transform.	8	
5	Algebraic Structures and Morphism: Algebraic Structures with one Binary Operation, Semi Groups, Monoids, Groups, Congruence Relation and Quotient Structures, Permutation Groups, Normal Subgroups, Quotient group, Homomorphism & Isomorphism (Elementary properties only). Algebraic Structures with two Binary Operation, Rings, Integral Domain and Fields. Boolean algebra and Boolean Ring, Identities of Boolean Algebra, Duality, Representation of Boolean Function	8	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Recognize the concept of the terms span, linear independence, basis and dimension and apply these concepts to various vector spaces and subspaces.		
CO 2	Solve problems involving recurrence relations and generating functions & also counting techniques and combinatorics in the context of discrete probability.		
CO 3	Learn the ideas of probability, random variable and distribution and their application in engineering environment.		
CO 4	Write periodic function in terms of sine and cosine terms in Fourier series and also to get knowledge in Fourier transforms.		

CO 5	Solve problems using counting techniques and combinatorics in the context of discrete probability.
CO6	Classify the algebraic structure for a given mathematical problem and evaluate Boolean functions and simplify expressions using the properties of Boolean algebra
Learning Resources:	
1.	C L Liu and D P Mohapatra, Elements of Discrete Mathematics A Computer Oriented Approach, 3rd Edition by, Tata McGraw – Hill.
2.	N. Chandrasekaran and M. Uma Parvathi, Discrete Mathematics, PHI.
3.	Loknath Debnath–Integral transforms and their applications.
4.	B. Chakraborty and M.K. Sen–Discrete mathematics.
5.	Koshy–Discrete Mathematics and Application
6.	Jyoti Medhi–Stochastic Process.
7.	S.K. Mapa –Abstract and Linear Algebra.
8.	Kenneth H. Rosen, Discrete Mathematics and its Applications, Tata McGraw – Hill
9.	Susanna S. Epp, Discrete Mathematics with Applications,4th edition, Wadsworth Publishing Co. Inc.
10.	Douglas Brent West, Introduction to Graph Theory, Prentice Hall.
11.	B. Davies-Integral transforms and their applications.
12.	Friedberg, Insel, Spence–Linear Algebra
13.	Clark John, Holton Derek Allan, A First Look at Graph Theory, World Scientific.

Name of the Course	Advanced Data Structure
Course Code: PGCSPC102	Semester: 1st
Duration: 6 months	Total Marks: 100
Teaching Scheme	Examination Scheme
Theory Contact Hrs.: 3 hrs./week	Mid Semester 1 Exam: 15 Marks
Tutorial Contact Hrs.: 0 hrs./week	Mid Semester 2 Exam: 15 Marks
Practical : 0 hrs./week	Other Assessment tools (Assignment, Quiz etc.): 20 Marks
Credit point: 3	End Semester Exam:100 Marks (To be mapped into 50 mark)

Objective:

1. To understand the concept of time and space complexity
2. To implement balanced trees
3. To implement various priority queues
4. To analyse various advance data structures such as hashing, tries and suffix trees

Pre-Requisite

1. Data Structure, Algorithm

Detailed Syllabus

Module	Content	Hrs	Marks
1	Review of Fundamental Data Structures Recap of basic data structures (arrays, linked lists, stacks, queues) Analysis of time and space complexity Big O notation	4	
2	Advanced Tree Structures Binary search trees (BSTs), AVL trees Red-black trees, B-trees and B*-trees for efficient indexing and searching. Van Emde Boas trees for efficient predecessor/successor search operations	10	
3	Heaps and Priority Queues, Min-heaps and max-heaps, Priority queue operations (insert, delete, extract-min/max) Fibonacci heaps	6	
4	Disjoint set data structure, Union find using array, linked list, union by rank, union by rank and path compression	8	
5	Advanced Techniques, Hashing and collision resolution, Tries and Patricia trees, Suffix trees	8	

Course outcomes

After completion of the course, a student would be able to:

- CO 1** **Compute** complexities of data structures and algorithms for a given problem.
- CO 2** **Implement** various balanced tree data structures.
- CO 3** **Implement** various Priority Queue data structures.
- CO 4** **Apply** disjoint set data structure
- CO 5** **Design** algorithms using Hashing, Tries and Suffix trees.

Learning Resources:

1. “Introduction to Algorithms,3rd edition”, T.H.Cormen, C.E. Leiserson, R.L.Rivest and C. Stein , PHI
2. Jon Kleinberg and Éva Tardos, Algorithm Design, Pearson, 2005.
3. Skiena, Steven S. The algorithm design manual. Vol. 2. New York: springer, 1998.

Name of the Course		Machine Learning	
Course Code: PGCSPC103		Semester: 1 st	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: 3 hrs./week		Mid Semester 1 Exam: 15 Marks	
Tutorial Contact Hrs.: 0 hrs./week		Mid Semester 2 Exam: 15 Marks	
Practical: 0 hrs./week		Other Assessment tools (Assignment, Quiz etc.): 20 Marks	
Credit Points: 3		End Semester Exam:100 Marks (To be mapped into 50 mark)	
Objective:			
1.	Develop an appreciation for what is involved in Learning models from data		
2.	Understand a wide variety of learning algorithms		
3.	Understand how to evaluate models generated from data		
4.	Apply the algorithms to a real problem, optimize the models learned and report on the expected		
Pre-Requisite			
1.	Mathematics		
Detailed Syllabus			
Module	Content	Hrs	Marks
1	Introduction: Prepare the Data for Machine Learning Algorithms, Data Cleaning, Handling Text and Categorical Attributes, Handling Missing Values, Exploration of Data using Visualization, Types of Machine Learning Systems.	5	
2	Linear Regression: Linear regression, Gradient Descent Algorithm for Linear Regression Model, Polynomial model, Regularization, Multi-Collinearity.	6	
3	Classification: Training a Binary Classifier, Measuring Performance, Logistic Regression, Multiclass Classifier, Multi-label Classification, Multi-output Classification	7	
4	Some Supervised Machine Learning Algorithms: k-Nearest Neighbours (KNN), Naive Bayes Classifiers, Decision Trees, Ensembles of Decision Trees: Random Forests, Support Vector Machines, Model Evaluation and Improvement	6	
5	Dimensionality Reduction: Dimensionality Reduction, Feature Extraction, and Manifold Learning, Principal Component Analysis (PCA), Kernel PCA, Selecting a Kernel and Tuning Hyper-parameters, Other Dimensionality Reduction Techniques	6	
6	Unsupervised Learning: Clustering: K-Means, Image Segmentation using clustering, Creating Product Segments Using Clustering, Finding Optimal Number of Clusters Using Elbow Curve Method, Normalizing the Features, Hierarchical Clustering, Compare the Clusters Created by K-Means and Hierarchical Clustering, Anomaly Detection using Gaussian Mixtures, Assessment Metrics for Clustering Algorithms.	6	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Describe the concepts of Machine Learning.		
CO 2	Implement algorithms of Machine Learning.		
CO 3	Develop Machine Learning models		
CO 4	Apply Machine Learning Models to classification and recognition problems.		
CO 5	Design various Machine Learning algorithms for real-world applications for model optimization.		
Learning Resources:			
1.	Christopher Bishop. Pattern Recognition and Machine Learning. 2e		
2.	Machine Learning by Tom Mitchell, McGraw Hill Education		
3.	Devi V.S.; Murty, M.N. (2011) Pattern Recognition: An Introduction, Universities Press, Hyderabad.		
4.	R. O. Duda, P. E. Hart and D. G. Stork, Pattern Classification, Wiley, 2000.		

Name of the Course		Artificial Intelligence	
Course Code: PGCSPC104		Semester: 1 st	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: 3 hrs./week		Mid Semester 1 Exam: 15 Marks	
Tutorial Contact Hrs.: 0 hrs./week		Mid Semester 2 Exam: 15 Marks	
Practical: 0 hrs./week		Other Assessment tools (Assignment, Quiz etc.): 20 Marks	
Credit Points: 3		End Semester Exam:100 Marks (To be mapped into 50 mark)	
Objective:			
1.	The main purpose of this course is to provide the most fundamental knowledge to the students so that they can understand what the AI is.		
2.	Apply the basic principles, models and algorithms of AI to recognize model and solve problems in the analysis and design of information systems.		
3.	Analyse the structures and algorithms of a selection of techniques related to searching, reasoning and language processing.		
Pre-Requisite			
1.	Strong knowledge of Mathematics.		
2.	Good command over programming languages		
3.	Design & Analysis of Algorithm		
Detailed Syllabus			
Module	Content	Hr s	Marks
1	Introduction of AI and Agents: Overview of Artificial Intelligence, Problems of AI, AI technique, Nature of Environment, Intelligent Agents, Structure of Agents, goal based agents, utility based agents, learning agents, Problem solving agents. Knowledge Based Agents.	5	
2	Problem Solving: Problem Space &Search: Defining the problem as state space search, production system, problem characteristics, issues in the design of search programs.	3	
3	Search Techniques: Searching for solutions; uniform search strategies: breadth first search, depth first search, depth limited search, bidirectional search, comparing uniform search strategies. Greedy best-first search, A* search, heuristic search, Hill climbing search, simulated annealing search, the minimax search procedure, alpha-beta pruning, constraint satisfaction problems, Games, optimal decisions & strategies in games.	10	
4	Knowledge &Reasoning and Representing Knowledge using Rules: Knowledge representation issues, representation & mapping, approaches to knowledge representation, The First Order Predicate Logic, Semantic Nets, Frames and Scripts Formalisms, Resolution in Predicate Logic, Unification, Strategies for Resolution by Refutation, Procedural verses declarative knowledge, logic programming, forward verses backward reasoning, matching, control knowledge. Automated Reasoning.	7	
5	Planning and Learning: Overview, components of a planning system, Goal stack planning, Hierarchical planning, other planning techniques, partial order planning, planning graphs, planning with propositional logic. Forms of learning, inductive learning, Learning from observations learning decision trees, learning using relevance information, neural net learning & genetic learning. Reinforcement Learning, Perception,	5	
6.	Natural Language Processing, Expert Systems and others: Introduction, Syntactic processing, semantic analysis, discourse & pragmatic processing, expert system shells, knowledge acquisition, Artificial Neural Networks, Human-Computer Interaction, Hidden Markov Models, Rule-based systems, AI-Based Programming Tools.	6	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Analyse different types of AI techniques and Agents		
CO 2	Estimate the problems of Space and Search		
CO 3	Apply different search techniques that play an important role in AI		
CO 4	Determine the basic issues of knowledge representation and reasoning		
CO 5	Implement different planning system and Learning System.		
CO6	Apply Natural Language Processing, Expert Systems and AI-Based Programming Tools		
Learning Resources:			
1.	Introduction to Artificial Intelligence, E. Charniak, et.al., PEARSON		
2.	Artificial Intelligence, P. H. Winston, PEARSON		
3.	Artificial Intelligence, E. Rich and K. Knight, TMH.		
4.	“Artificial Intelligence: A Modern Approach” Stuart Russell, Peter Norvig,2nd Edition, Prentice Hall		

5.	Artificial Intelligence and Expert Systems – Dan W. Patterson PHI
6.	Expert Systems: Principles and Programming- Fourth Edn, Giarrantana/ Riley, Thomson
7.	Artificial Neural Networks , B. Yagna Narayana, PHI
8.	The Handbook of Artificial Intelligence, Vol.1,2 and 3, Kaufman Inc.
9.	Logic & Prolog Programming, Saroj Kaushik, New Age International
10.	PROLOG Programming for Artificial Intelligence. Ivan Bratka- Third Edition – Pearson Education

Name of the Course	Universal Human Values (UHV)
Course Code: PGCSHS101	Semester: 1st
Duration: 6 months	Total Marks: 100
Teaching Scheme	Examination Scheme
Theory Contact Hrs.: 2 hrs./week	Mid Semester 1 Exam: 15 Marks
Tutorial Contact Hrs.: 0 hrs./week	Mid Semester 2 Exam: 15 Marks
Practical: 0 hrs./week	Other Assessment tools (Assignment, Quiz etc.): 20 Marks
Credit Points: 2	End Semester Exam:100 Marks (To be mapped into 50 mark)

Objective:

1.	To help the students appreciate the essential complementarity between ‘VALUES’ and ‘SKILLS’ to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2.	To facilitate the development of a holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of existence. Such a holistic perspective forms the basis of Universal Human Values and movement towards value-based living in a natural way.
3.	To highlight plausible implications of such a holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behaviour and mutually enriching interaction with nature. Thus, this course is intended to provide a much-needed orientation input in value education to the young enquiring minds.

Pre-Requisite

1.	Nil
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Detailed Syllabus

Module	Content	Hrs	Marks
1	Introduction: Introduction to value education, Engineering ethics: Social responsibilities of an engineer.	6	
2	Harmony in the human being: Self exploration, needs of the self and body	6	
3	The basic need of a healthy society to extend the family: Understanding harmony between family and society, understanding values in human relationships. Trust and Respect as the foundational values	9	
4	Develop the right understanding of the harmony at all levels: Understanding harmony in the nature and existence holistic perception of harmony at all levels of existence include practice session to discuss human being as cause of imbalance in nature.	8	
5	Implication of the above Holistic Understanding of Harmony on Professional Ethics: The basis of ethical human conduct. Human rights violation and social disparities issues in professional ethics: value-based life and profession.	7	

Course outcomes

After completion of the course, a student would be able to:

CO 1	Evaluate the significance of value inputs in formal education and start applying them in their life and professions.
CO 2	Distinguish between values and skills, real happiness and accumulation of physical facilities, the self and the Body, Intension and competence of an individual etc.
CO 3	Analyse the value of harmonious relationship based on trust and respect in their life and profession.
CO 4	Examine the role of a human being in ensuring harmony in society and nature.
CO 5	Apply the Understanding of ethical conduct to formulate the strategy for ethical life and profession.

Learning Resources:

1.	A.N. Tripathi, new age international publishers: 2003- Human values. Human Society in Ethics 2 polities- Bertrand Russell.
2.	Small is beautiful – E.F. Schumacher philosophy of humanism: Corliss Layout. Foundation Course in value Education: Gour.R.R./sangal R.

PRACTICAL

Name of the Course		Advanced data structure lab	
Course Code: PGCSPCL105		Semester: 1 st	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: Nil		Continuous Internal Evaluation: 60	
Tutorial Contact Hrs.: Nil		Semester End Practical Exam: 20	
Practical: 3 hrs./week		External Viva Voce: 20	
Credit Points: 1.5			
Detailed Syllabus			
Module	Content	Hrs	Marks
1	To perform various operations on AVL trees.	3	
2	To perform various operations on B-Trees and B ⁺ trees	3	
3	To perform various operations on Red Black trees.	3	
4	To implement and compare performance of various priority queues	3	
5	To implement disjoint set data structure and compare the performance based on underlying methodology used.	6	
6.	To implement Hashing and analyse performance of different collision resolution techniques.	3	
7.	To implement Tries and Patricia Tries	6	
8.	To implement Suffix trees	3	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Implement various balanced search trees		
CO 2	Compare performance of different priority queue implementations		
CO 3	Implement Disjoint set data structure		
CO 4	Analyse hashing techniques		
CO 5	Implement tries and suffix trees		

Name of the Course		Machine learning lab	
Course Code: PGCSPCL106		Semester: 1 st	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: Nil		Continuous Internal Evaluation: 60	
Tutorial Contact Hrs.: Nil		Semester End Practical Exam: 20	
Practical: 3 hrs./week		External Viva Voce: 20	
Credit Points: 1.5			
Detailed Syllabus			
Module	Content	Hrs	Marks
1	Implement K-nearest neighbours classification using python.	3	
2	Implement linear regression using python.	3	
3	Implement Naive Bayes theorem to classify the English text using python	6	
4	Implement K-means clustering using python.	3	
5	Write a program to demonstrate the working of decision tree.	3	
6.	Implement K-nearest neighbours classification using python.	3	
7.	Implement linear regression using python.	3	
8.	Implement Naive Bayes theorem to classify the English text using python	6	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Design and implement machine learning solutions to classification, regression problems.		
CO 2	Apply appropriate data sets to the Machine Learning algorithms		
CO 3	Identify and apply Machine Learning algorithms to solve real world problems		
CO 4	Apply supervised and unsupervised techniques on various data sets.		
CO 5	Evaluate the machine learning models pre-processed through various feature engineering algorithms by python programming		
CO6	Analyze the complexity of Machine Learning algorithms and their limitations		

2 nd SEM							
THEORY							
SL. NO.	PAPER CODE	PAPER NAME	L	T	P	CONTACT HRS./WEEK	CREDIT
01	PGCSPC207	Deep Learning	3	0	0	3	3
02	PGCSPC208	Advanced Algorithm	3	0	0	3	3
03	PGCSPC209	Information Security	3	0	0	3	3
03	PGRES201	Research Methodology and IPR	2	0	0	2	2
05	PGCSPEC201	A. Soft Computing B. Data Science	3	0	0	3	3
06	PGCSPEC202	A. Advanced Software Engineering B. Robotics	3	0	0	3	3
PRACTICAL							
01	PGCSPCL210	Deep Learning Lab	0	0	3	3	1.5
02	PGCSPCL211	Advanced Algorithm Lab	0	0	3	3	1.5
02	PGCSSEM201	Assignment/seminar	0	0	4	4	2
		Total	17	0	10	27	22

Name of the Course	Deep Learning
Course Code: PGCSPC207	Semester: 2nd
Duration: 6 months	Total Marks: 100
Teaching Scheme	Examination Scheme
Theory Contact Hrs.: 3 hrs./week	Mid Semester 1 Exam: 15 Marks
Tutorial Contact Hrs.: 0 hrs./week	Mid Semester 2 Exam: 15 Marks
Practical: 0 hrs./week	Other Assessment tools (Assignment, Quiz etc.): 20 Marks
Credit Points: 3	End Semester Exam:100 Marks (To be mapped into 50 mark)

Objective:

- Understand the concepts of Tensor Flow, Keras, its main functions, operations and execution.
- Implement deep learning algorithms, understand neural networks and traverse the layers of data abstraction which will empower the student to understand data more precisely.
- Build deep learning models in Tensor Flow and interpret the results.
- Learn topics such as Convolutional neural networks, recurrent neural networks, LSTM, GRU, training deep networks and high-level interfaces.
- Understand the Auto encoders, GAN and Reinforcement learning concepts.

Pre-Requisite

- Machine learning

Detailed Syllabus

Module	Content	Hrs	Marks
1	Artificial Neural Networks (ANNs), Perceptron, Multi-Layer Perceptron (MLP), Back propagation, Hyper-parameter selection, Activation functions.	10	
2	Training Deep Neural Networks, Vanishing Gradients Problems, Batch Normalization, Optimizers: Ada Grad, RMS Prop, Adam Optimization, Regularization.	6	
3	Convolutional Neural Networks (CNNs), Convolutional Layers, Filters, Pooling strategies, CNN Architectures, Reutilizing Pretrained Layers	6	
4	Recurrent Neural Networks (RNNs), Recurrent Neurons, Training RNNs, LSTM, GRU.	5	
5	Auto encoders and Generative Adversarial Networks (GAN)	4	
6	Reinforcement Learning, Introduction to Open AI Gym, Markov Decision Processes, Q-Learning.	5	

Course outcomes

After completion of the course, a student would be able to:

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|-------------|---|
| CO 1 | Describe the concepts of Tensor Flow, Keras, its main functions, operations and execution. |
| CO 2 | Explain deep learning algorithms and neural networks. |
| CO 3 | Develop models of convolutional neural networks (CNN), recurrent neural networks (RNN), LSTM, GRU, training deep networks and high-level interfaces. |
| CO 4 | Apply Deep Learning Models to realise the concepts of Auto encoders and GAN. |
| CO 5 | Design Deep Learning algorithms for Reinforcement learning. |

Learning Resources:

- Christopher Bishop. Pattern Recognition and Machine Learning. 2e
- Machine Learning by Tom Mitchell, McGraw Hill Education
- Devi V.S.; Murty, M.N. (2011) Pattern Recognition: An Introduction, Universities Press, Hyderabad.
- R. O. Duda, P. E. Hart and D. G. Stork, Pattern Classification, Wiley, 2000.
- Good fellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press, 2016
- Christopher Bishop. Pattern Recognition and Machine Learning. 2e

Name of the Course	Advanced Algorithm
Course Code: PGCSPC208	Semester: 2nd
Duration: 6 months	Total Marks: 100
Teaching Scheme	Examination Scheme
Theory Contact Hrs.: 3 hrs./week	Mid Semester 1 Exam: 15 Marks
Tutorial Contact Hrs.: 0 hrs./week	Mid Semester 2 Exam: 15 Marks
Practical: 0 hrs./week	Other Assessment tools (Assignment, Quiz etc.): 20 Marks
Credit Points: 3	End Semester Exam:100 Marks (To be mapped into 50 mark)

Objective:

- To understand the concept of randomized algorithms and NP completeness
- To implement graph theory algorithms
- To able to use parametric algorithms in real life practical problems.
- To understand the concept of approximate algorithms

Pre-Requisite

- Algorithm

Detailed Syllabus			
Module	Content	Hrs	Marks
1	Amortized Analysis	3	
2	NP Completeness: The Class P, Examples of problems in P, The Class NP, Examples of problems in NP, P versus NP, , Polynomial time reducibility, NP-completeness, The Cook–Levin Theorem.	5	
3	Network Flows, Ford-Fulkerson Algorithm, Edmond-Karp Algorithm .Max-Flow Min-Cut Theorem, Bipartite Matching, stable matching	12	
4	Probabilistic Analysis and Randomized algorithms, The hiring Problem, Indicator random variables, Randomized algorithms.	8	
5	Approximate Algorithms, The vertex cover problem, The Travelling Salesman Problem, The set covering Problem,	8	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Analyze Randomized algorithms for a given problem.		
CO 2	Analyze problems to decide NP completeness		
CO 3	Apply graph algorithms to real life problems		
CO 4	Apply Amortized analysis to various algorithms		
CO 5	Design approximate algorithms for problems.		
Learning Resources:			
1.	“Introduction to Algorithms, 3 rd edition”, T.H.Cormen, C.E. Leiserson, R.L.Rivest and C. Stein , PHI		
2.	Randomized •algorithms, Rajeev Motwani, PrabhakarRaghavan, Cambridge University Press		
3.	Skiena, Steven S. The algorithm design manual. Vol. 2. New York: springer, 1998.		
4.	Approximation Algorithms, Vazirani, Vijay V, 2003, Springer.		

Name of the Course		Information Security	
Course Code: PGCSPC209		Semester: 2nd	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: 3 hrs./week		Mid Semester 1 Exam: 15 Marks	
Tutorial Contact Hrs.: 0 hrs./week		Mid Semester 2 Exam: 15 Marks	
Practical: 0 hrs./week		Other Assessment tools (Assignment, Quiz etc.): 20 Marks	
Credit Points: 3		End Semester Exam:100 Marks (To be mapped into 50 mark)	
Objective:			
1.	To realize the importance of data security practices in the aspect of critical data transmission		
2.	To explain some of the common cryptographic and data hiding solutions with their effectiveness		
3.	To illustrate a practical approach of authentication in the aspect of secret image sharing		
4.	To plan some of the security solutions for user authentication and sensitive data validation		
Pre-Requisite			
1.	Computer Network		
Detailed Syllabus			
Module	Content	Hrs	Marks
1	Introductory Concepts on Information Security: Need for information security, security models, major principles of security- authentication, confidentiality, integrity, non-repudiation, access control and availability, classification of different attacks- application-level and network-level attacks, malicious software based attacks, packet sniffing, packet spoofing	6	
2	Symmetric Key Cryptographic Concepts: plaintext & cipher text, issues of key ranges and sizes, Diffie-Helman Key exchange algorithm, man in the middle attack, idea of cryptanalysis, cipher text formation with substitution & transposition techniques, idea of stream & block cipher, symmetric key encryption algorithm- DES, double & triple DES, AES	10	
3	Asymmetric Key Cryptographic Concepts: Basic idea, comparison of both symmetric and asymmetric key cryptography, asymmetric key encryption with RSA, concept of digital signature with message digest or hashing, combining both symmetric and asymmetric key encryption concept for secure data transmission.	6	
4	Data Hiding Concepts for Security: Basic idea, use of steganography, combining cryptographic and stenographic concepts for security, stenographic techniques- spatial and transform domain with their comparisons and examples, concept of watermarking and their classifications, idea of multi-	8	

	watermarking, attacks related to watermarking, data hiding quality matrices- PSNR, MSE, Image Fidelity, CC etc.		
5	User Authentication Mechanism: password based authentication and its related issues, random challenge based authentication, certificate based authentication, idea of visual cryptography- share generation concept in visual cryptography, use of visual cryptographic concept with image steganography for digital signature implementation	6	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Understand the scope of information security in the context of various attacks		
CO 2	Use Symmetric Key cryptography for secret data communication		
CO 3	Explain the role of asymmetric key cryptography for secure and trusted data communication		
CO 4	Illustrate data hiding mechanism for authentication and secure data transmission		
CO 5	Outline some data security solutions in the context of both user and document validations		
Learning Resources:			
1.	William Stallings, Cryptography and Network Security Principles and Practices, 5th Edition, Prentice Hall		
2.	C. Kaufman, R. Perlman and M. Speciner, Network Security: Private communication, 2nd Edition, Pearson Education		
3.	Atul Kahate, Cryptography & Network Security, 3rd Edition, McGraw Hill Education (India) Private Limited		
4.	Merike Kaeo, Designing Network Security, 2nd Edition, Pearson Books		
5.	Information Hiding: Steganography and Watermarking: attacks and countermeasures, Neil F. Johnson, Zoran Duric, Sushil Jajodia, Springer Science & Business Media, 2001		
6.	Digital Watermarking and Steganography: Fundamentals and Techniques, Frank Y. Shih, Second Edition, CRC Press, 2017		

Name of the Course		Research Methodology and IPR	
Course Code: PGRES201		Semester: 2 nd	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: 2 hrs./week		Mid Semester 1 Exam: 15 Marks	
Tutorial Contact Hrs.: 0 hrs./week		Mid Semester 2 Exam: 15 Marks	
Practical: 0 hrs./week		Other Assessment tools (Assignment, Quiz etc.): 20 Marks	
Credit Points: 2		End Semester Exam:100 Marks (To be mapped into 50 mark)	
Objective:			
1.	To learn the features of Research Methodology		
2.	To identify the rules of Copy Right, Design and Trade Mark		
3.	To analyse Intellectual Property Rights		
4.	To apply for a Patent and defend Patent Prosecution		
Pre-Requisite			
1.	Nil		
Detailed Syllabus			
Module	Content	Hrs	Marks
1	Course-based degree and research degree, selection of research topic, the role of supervisor, Academic research and developmental research. Keyword based search and citation based search, cite score and impact factor. Literature survey using Web of Science and Scopus. Database search using Science Direct, Springer Link, Research Gate, Academia.edu, Google Scholar etc. Reference Management Software: Jabref, EndNote, Zotero etc.	10	
2	Investigation and generation of data. First principles model and empirical model. Features and constraints of Technical Presentation. Standard table, graph and image. Technical presentation vs journal paper. Unethical activity: data fabrication and plagiarism.	6	
3	Discrete Probability Distribution and Continuous Probability Distribution. Probability Density Function, Mean and Variance of Probability Density Function. Normal Distribution and Central Limit Theorem. Testing of Hypothesis, Errors in decision making.	6	
4	Definition of IPR, types of Intellectual Property, meaning of “right” in IPR, meaning of “property” in IPR, meaning of “intellectual” in IPR, infringement in IPR, IPR and its duration, Invention and patent, rights of patentee. Process for grant of patent: abstract, description, specification, claims for invention. Filing of Application: Self application or application through Patent Agent. Types of Application: Provisional and Complete	8	

	Application, Convention Application, International Application under Patent Cooperation Treaty (PCT),		
5	Copyright, its features and duration, Design Act, its feature, duration and Hague Agreement. Trade Marks Act, its feature and duration, International application for Trade Mark World Intellectual Property Organization (WIPO), European Patent Office (EPO) Database, Berne Convention, International Bureau (IB) of WIPO.	6	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Explain the techniques Literature Survey		
CO 2	Interpret Data Modelling and Technical Presentation		
CO 3	Examine the concept of Design of Experiments		
CO 4	Apply for a Patent in the context of IPR		
CO 5	Investigate different Responsible Organizations for IPR and Database		
Learning Resources:			
1.	Important Guidelines on Research Methodology by R. S. Agarwal, published by B. R. Publishing Corporation		
2.	Research Methodology by Suresh Chandra, published by Narosa Publishing House		
3.	Research Methodology by A. K. Ohdedar, published by Bengal Library Association		
4.	Text Book of Intellectual Property Rights by N. K. Acharya, published by Asia Law House		
5.	Law Relating to Intellectual Property Rights by V. K. Ahuja, published by Lexis Nexis		

Name of the Course		Soft Computing	
Course Code: PGCSPEC201A		Semester: 2 nd	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: 3 hrs./week		Mid Semester 1 Exam: 15 Marks	
Tutorial Contact Hrs.: 0 hrs./week		Mid Semester 2 Exam: 15 Marks	
Practical: 0 hrs./week		Other Assessment tools (Assignment, Quiz etc.): 20 Marks	
Credit Points: 3		End Semester Exam:100 Marks (To be mapped into 50 mark)	
Objective:			
1.	To understand basic soft computing techniques		
2.	To learn how to use soft computing technique for a particular problem		
3.	To implement hybrid soft computing techniques		
Pre-Requisite			
1.	Discrete Mathematics		
2.	Design and Analysis of Algorithm		
Detailed Syllabus			
Module	Content	Hrs	Marks
1	Introduction: Concept of computing systems, Characteristics of Soft computing, some applications of Soft computing techniques, Different learning methods: Supervised, Unsupervised & reinforced; Simple Clustering algorithm, k-means & k-medoid based algorithm.	4	
2	Artificial Neural Network: Basic concept of neural networks, Mathematical model, Typical architectures: single layer, multilayer, Common activation functions; basic models, Perceptron, Multilayer feed forward network, Back propagation, ADALINE, MADALINE, Different issues regarding convergence of Multilayer Perceptron, Self-Organizing Feature Maps.	8	
3	Fuzzy Logic: Fuzzy Sets, Basic Definitions and Terminology, membership function, Set-theoretic operation. Fuzzy union, intersection and complement, various T-norm and T-conorm operators, Fuzzy Relations. Fuzzy Logic, Approximate Reasoning, Compositional Rule of Inference.	5	
4	Evolutionary Algorithms: Genetic Algorithms: Simple GA, Encoding Techniques, Crossover, mutation, inversion and deletion, Multi-objective Genetic Algorithm (MOGA). Applications of Genetic Algorithm: genetic algorithms in search and optimization, GA based clustering Algorithm. Other Soft Computing techniques: Simulated Annealing, Tabu search, Ant colony optimization (ACO), Particle Swarm Optimization (PSO).	12	
5	Hybrid Soft Computing Techniques: Neuro-fuzzy hybrid algorithm, genetic neuro hybrid algorithm, fuzzy-genetic hybrid algorithm, genetic-fuzzy hybrid algorithm, GA-PSO hybrid algorithm.	7	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Understand the concept of soft computing		
CO 2	Explain fuzzy sets and represent these sets by membership functions		

CO 3	Compare the relation between real brains and simple artificial neural network models
CO 4	Analyze Evolutionary algorithms for single and multiple objective optimization problem
CO 5	Design Neuro fuzzy and other hybrid approaches of soft computing techniques for problem solving
Learning Resources:	
1.	“Neuro-Fuzzy and Soft computing”, Jang, Sun, Mizutani, PHI
2.	“Neural Networks: A Comprehensive Foundation”, Simon Haykin, Prentice Hall.
3.	“Genetic Algorithms in search, Optimization & Machine Learning”, David E. Goldberg, Pearson/PHI
4.	“Fuzzy Sets and Fuzzy Logic: Theory and Applications”, George J. Klir and Bo Yuan, Prentice Hall
5.	S. Rajasekaran and G.A.V.Pai, “Neural Networks, Fuzzy Logic and Genetic Algorithms”, PHI
6.	Fuzzy logic with engineering applications, Timothy J. Ross, John Wiley and Sons.
7.	D. Ruan, Intelligent Hybrid Systems, Kluwer Academic Publisher, 1997.

Name of the Course	Data Science
Course Code: PGCSPEC201B	Semester: 2nd
Duration: 6 months	Total Marks: 100
Teaching Scheme	Examination Scheme
Theory Contact Hrs.: 3 hrs./week	Mid Semester 1 Exam: 15 Marks
Tutorial Contact Hrs.: 0 hrs./week	Mid Semester 2 Exam: 15 Marks
Practical: 0 hrs./week	Other Assessment tools (Assignment, Quiz etc.): 20 Marks
Credit Points: 3	End Semester Exam:100 Marks (To be mapped into 50 mark)

Objective:	
1.	The course is aimed at Building the fundamentals of data science.
2.	Imparting design thinking capability to build big-data
3.	Developing design skills of models for big data problems
	Gaining practical experience in programming tools for data sciences
	Empowering students with tools and techniques used in data science

Pre-Requisite	
1.	DBMS and e knowledge of one Programming Language (Java preferably), Practice of SQL (queries and sub queries), exposure to Linux Environment

Detailed Syllabus			
Module	Content	Hrs	Marks
1	Introduction: Big Data and Data Science - Big Data Analytics, Business intelligence vs Big data, big data frameworks, Current landscape of analytics, data visualisation techniques, visualization software.	4	
2	EDA Exploratory Data Analysis: Exploratory Data Analysis (EDA), statistical measures, Basic tools (plots, graphs and summary statistics) of EDA, Data Analytics Lifecycle, Discovery	5	
3	Basic Statistical Inference: Developing Initial Hypotheses, Identifying Potential Data Sources, EDA case study, testing hypotheses on means, proportions and variances.	5	
4	Regression models: Regression models: Simple linear regression, least-squares principle, MLR, Multiple correlation, Partial correlation	5	
5	Linear Algebra Basics: Matrices to represent relations between data, Linear algebraic operations on matrices, Matrix decomposition: Singular Value Decomposition (SVD) and Principal Component Analysis (PCA).	5	
	Data Pre-processing and Feature Selection : Data cleaning, Data integration, Data Reduction, Data Transformation and Data Discretization, Feature Generation and Feature Selection, Feature Selection algorithms: Filters, Wrappers, Embedded	6	
	Basic Machine Learning Algorithms: Classifiers, Decision tree, Naive Bayes, k-Nearest Neighbours (KNN), SVM, Ensemble methods, Random Forests, logistic regression, k-means, Association Rule mining	6	

Course outcomes	
After completion of the course, a student would be able to:	
CO 1	Apply data visualization in big-data analytics
CO 2	Utilise EDA, inference and regression techniques
CO 3	Utilize Matrix decomposition techniques to perform data analysis
CO 4	Apply data pre-processing techniques
CO 5	Apply Basic Machine Learning Algorithms

Learning Resources:	
1.	Mining of Massive Datasets. v2.1, Jure Leskovek, Anand Rajaraman and Jeffrey Ullman., Cambridge University Press. (2019). (free online)
2.	Big Data Analytics, paperback 2nd ed., Seema Acharya, Subhasini Chellappan, Wiley (2019)

3.	Doing Data Science, Straight Talk From The Frontline, Cathy O'Neil and Rachel Schutt, O'Reilly.
4.	Data Mining: Concepts and Techniques”, Third Edition, 2 Jiawei Han, Micheline)
5.	Kamber and Jian Pei, ISBN 0123814790. Big Data and Business Analytics, Jay Liebowitz, CRC press
6.	Data mining methods, 24th edition, C. Rajan, Narosa

Name of the Course	Advanced Software Engineering
Course Code: PGCSPEC202A	Semester: 2nd
Duration: 6 months	Total Marks: 100
Teaching Scheme	Examination Scheme
Theory Contact Hrs.: 3 hrs./week	Mid Semester 1 Exam: 15 Marks
Tutorial Contact Hrs.: 0 hrs./week	Mid Semester 2 Exam: 15 Marks
Practical: 0 hrs./week	Other Assessment tools (Assignment, Quiz etc.): 20 Marks
Credit Points: 3	End Semester Exam:100 Marks (To be mapped into 50 mark)

Objective:	
1.	To learn the different models for the development of a software product
2.	To explore the designing, coding and testing to develop software product
3.	To asses quality of software product to sustain in the market

Pre-Requisite	
1.	Fundamentals of Programming knowledge, C, C++ etc.

Detailed Syllabus			
Module	Content	Hrs	Marks
1	Software Development Process Models: Waterfall, Spiral, Prototyping, RAD, Evolutionary, Software Requirement and Feasibility Analysis, Cost- Benefit Analysis, etc.	5	
2	Software Design: Context Diagram, DFD, Data Dictionary, ER diagram, Decision Tree, Decision Table, Structured Chart, Structured English, Top-Down and Bottom-Up design, Modular Programming, Module Relationship- Coupling, Cohesion, Functional vs. Object-Oriented approach, Design for Mobility, Pattern Based Design.	5	
3	Object Oriented Software Designing with UML: Basic idea of Object Oriented Modelling & its purpose, Primary goals of UML, static, dynamic and functional aspects of UML, purpose of different diagrams in UML- class, object and interface diagram, class association, dependencies, multiplicity of association, aggregation, brief idea of use case diagram, sequence diagram, collaboration diagram, state chart and transition diagram, activity and deployment diagram, concept of packages,	7	
4	Software Quality: Quality Attributes, Total Quality Management, Software Quality Assurance and Quality Control, Reliability, Reliability Models, SEI CMM and ISO 9001. PSP and Six Sigma. Clean room Technique. Software reuse, Software Maintenance.	6	
5	Software Testing: Software Testing Strategy, Unit Testing, Integration Testing, System Testing, Debugging, White-Box Testing, Basis Path Testing, Control Structure Testing, Black-Box Testing, Verification, Validation, Cyclomatic Complexity.	5	
6	Software Project Management& Software Configuration Management: Software Project Management Concepts, Software Project Management Plan, Tools for Project Plan – WBS, PERT, GANTT, Project Scheduling & Monitoring, Staffing, Cost Estimation, COCOMO, Software Reengineering Process model, Software Configuration Management (SCM), SCM Repository.	8	

Course outcomes	
After completion of the course, a student would be able to:	
CO 1	Differentiate among different types of SDLC models.
CO 2	Assess the quality of software
CO 3	Examine various testing techniques
CO 4	Review the activity of software project management with CASE study

Learning Resources:	
1.	Software Engineering: A practitioner’s approach– Pressman (TMH)
2.	Software Engineering: Pankaj Jalote (Wiley-India)
3.	Software Engineering: Rajib Mall (PHI)
4.	Software Engineering: Agarwal and Agarwal, (PHI)
5.	Software Engineering: Sommerville, Pearson
6.	Fundamentals of Software Engineering – C. Ghezzi, M. Jazayeri, D. Mandrioli
7.	Software Engineering Martin L. Shooman,– TMH

Name of the Course		Robotics	
Course Code: PGCSPEC202B		Semester: 2 nd	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: 3 hrs./week		Mid Semester 1 Exam: 15 Marks	
Tutorial Contact Hrs.: 0 hrs./week		Mid Semester 2 Exam: 15 Marks	
Practical: 0 hrs./week		Other Assessment tools (Assignment, Quiz etc.): 20 Marks	
Credit Points: 3		End Semester Exam:100 Marks (To be mapped into 50 mark)	
Objective:			
1.	Building the fundamentals of Robotics		
2.	Imparting design thinking capability to build Robot		
3.	For modelling create a useful mathematical representation of a physical system		
4.	Gaining practical experience in programming tools and techniques used in Robotics for Robotics		
5.	Empowering students with the knowledge of different tools used in robotics		
6.	Develop control logics to make the system follow a given trajectory and react to unexpected obstacles.		
Pre-Requisite			
1.	Basic Mathematics and knowledge of Programming		
Detailed Syllabus			
Module	Content	Hrs	Marks
1	ROBOT BASICS Basic concepts, Need, Law, History, Anatomy, specifications. Robot configurations: Cartesian, cylinder, polar and articulate. Robot wrist mechanism, Precision and accuracy of robot.	5	
2	ROBOT ELEMENTS End effectors-Classification, Types of Mechanical (Hydraulic and Pneumatic) actuation, Basic concepts of Gripper design, Robot drive system Types, Position and velocity feedback devices, Robot joints and links, Motion interpolation.	7	
3	ROBOT KINEMATICS AND CONTROL Robot kinematics – Basics of direct and inverse kinematics, Robot trajectories, 2D and 3D Transformation-Scaling, Rotation, Translation Homogeneous transformation. Control of robot manipulators – Point to point, Continuous Path Control, Robot programming	7	
4	SYSTEM MODELLING AND STABILTY ANALYSIS Introduction to system modelling, application of Laplace Transform, Block diagrams, Time response of first and second order system, analysis of stability of a system, introduction to controllers	6	
5	ROBOT SENSORS Sensors in robot – Touch sensors-Tactile sensor, Proximity and range sensors. Force sensor, Light sensors, Pressure sensors, Introduction to Machine Vision and Artificial Intelligence.	6	
6	ROBOT APPLICATIONS Industrial applications of robots, Medical, Household, Entertainment, Space, Underwater, Defence, Disaster management. Applications, Micro and Nan robots, Future Applications	5	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Explain the basic elements of industrial robots		
CO 2	Analyze robot kinematics and robotic control methods.		
CO 3	Develop the concept of system modelling and stability.		
CO 4	Classify the various sensors used in robots.		
CO 5	Summarize various industrial and non-industrial applications of robots		
Learning Resources:			
1.	Mikell P. Groover, Mitchell Weiss, Roger N Nagel, Nicholas G Odrey, “Industrial Robotics Technology, Programming and Applications”, Tata –McGraw Hill Pub. Co., 2008.		
2.	Deb. S.R and Sankha Deb, "Robotics Technology and Flexible Automation", Tata McGraw Hill Publishing Company Limited, 2010.		
3.	Klafter. R.D, Chmielewski. T.A, and Noggin’s., “Robot Engineering: An Integrated Approach”, Prentice Hall of India Pvt. Ltd., 1994.		
4.	Fu. K.S, Gonzalez. R.C & Lee. C.S.G, “Robotics control, sensing, vision and intelligence”, Tata- McGraw Hill Pub. Co., 2008.		
5.	Yu. “Industrial Robotics”, MIR Publishers Moscow, 1985.		
6.	Peter Corke, Robotics, Vision and Control, Springer, 2011		
7.	Kelly, A, Mobile Robotics: Mathematics, Models, and Methods, Cambridge		
8.	Roland Siegwart, Illah R. Nourbakhsh and Davide Scaramuzza, Introduction to Autonomous Mobile Robots, Second Edition, MIT Press 2011		

PRACTICAL

Name of the Course		Deep learning lab	
Course Code: PGCSPCL210		Semester: 2 nd	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: Nil		Continuous Internal Evaluation: 60	
Tutorial Contact Hrs.: Nil		Semester End Practical Exam: 20	
Practical: 3 hrs./week		External Viva Voce: 20	
Credit Points: 1.5			
Detailed Syllabus			
Module	Content	Hrs	Marks
1	Implementing Feedforward neural networks with Keras and Tensor Flow.	3	
2	Building Image classification model with CNN architectures	6	
3	Object detection using Transfer Learning of CNN architectures	6	
4	Using Auto encoder to implement anomaly detection.	3	
5	Regression using Deep Neural network.	6	
6.	Classification using Deep neural network.	6	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Learn the Fundamental Principles of Deep Learning.		
CO 2	Identify the Deep Learning Algorithms for Various Types of Learning Tasks in various domains.		
CO 3	Implement Deep Learning Algorithms and Solve Real-world problems.		
CO 4	Understand the basic concepts of deep neural network model and design the same.		
CO 5	Apply basic data pre-processing and tuning techniques. (Cognitive Knowledge Level		

Name of the Course		Advanced Algorithm lab	
Course Code: PGCSPCL211		Semester: 2 nd	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: Nil		Continuous Internal Evaluation: 60	
Tutorial Contact Hrs.: Nil		Semester End Practical Exam: 20	
Practical: 3 hrs./week		External Viva Voce: 20	
Credit Points: 1.5			
Detailed Syllabus			
Module	Content	Hrs	Marks
1	Implement median order statistics using random partition function and compare the performance with classical partition function.	6	
2	Execute randomized quick sort and compare performance with fixed pivot quicksort.	3	
3	Implement network flow algorithms	6	
4	Implement matching algorithms	6	
5	Design approximate algorithms for problems	6	
6.	Use reduction to solve unknown problems using algorithms for known problems	3	
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Implement randomized algorithms		
CO 2	Analyze network flow algorithms		
CO 3	Implement Matching algorithms		
CO 4	Apply reductions to solve problems		
CO 5	Design Approximate algorithms for problems		

3rd SEM							
THEORY							
SL. NO.	PAPER CODE	PAPER NAME	L	T	P	CONTACT HRS./WEEK	CREDIT
01	PGCSPEC303	A. Big Data Analytics B. Natural Language Processing C. Internet of Things D. Distributed Database	3	0	0	3	3
PRACTICAL							
01	PGCSDIS301	Dissertation-I	0	0	26	26	13
		Total	3	0	26	29	16

Name of the Course		Big Data Analytics	
Course Code: PGCSPEC303A		Semester: 3 rd	
Duration: 6 months		Total Marks: 100	
Teaching Scheme		Examination Scheme	
Theory Contact Hrs.: 3 hrs./week		Mid Semester 1 Exam: 15 Marks	
Tutorial Contact Hrs.: 0 hrs./week		Mid Semester 2 Exam: 15 Marks	
Practical: 0 hrs./week		Other Assessment tools (Assignment, Quiz etc.): 20 Marks	
Credit Points: 3		End Semester Exam:100 Marks (To be mapped into 50 marks)	
Objective:			
1.	To study the basic technologies that forms the foundations of Big Data Analytics		
2.	Provide an overview of Apache Hadoop, HDFS Concepts and Interfacing with HDFS		
3.	To understand the specialized aspects of big data including big data application, and big data analytics.		
4.	Increasing operational efficiency by understanding where bottlenecks are and how to fix them.		
Pre-Requisite			
1.	Strong knowledge of DBMS		
2.	Good command over programming languages		
3.	Basic Mathematics and Statistics		
Detailed Syllabus			
Module	Content	Hrs	Marks
1	Introduction to Big Data: History of big data, Elements of big data, why big data, Unstructured data, Data Storage and Analysis, Characteristics of Big Data, Using big data in businesses, Challenges in Big Data Analytics.	5	
2	Technologies for handling Big Data: History and overview of Hadoop, Functioning of Hadoop, limitations of RDBMS, RDBMS versus Hadoop, Cloud Computing for big data, Distributed computing challenges, Analyzing Data with Hadoop, Hadoop Streaming, Use case of Hadoop, Processing data with Hadoop, Managing resources and applications with Hadoop YARN (Yet another Resource Negotiator).	3	
3	HDFS(Hadoop Distributed File System): Hadoop distributors, the Design of HDFS, HDFS Concepts, Command Line Interface, Apache Hadoop, Hadoop file system interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives, Hadoop I/O (Compression, Serialization, Avro and File-Based Data structures).	10	
4	Map Reduce: Anatomy of a Map Reduce Job Run, Failures, Job Scheduling, Shuffle and Sort, Task Execution, Map Reduce Types and Formats, Map Reduce Features.	7	
5	Hadoop Eco System: Pig (Introduction to Pig, Execution Modes of Pig, Comparison of Pig with Databases, Grunt, Pig Latin, User Defined Functions, Data Processing operators), Hive (Hive Shell, Hive Services, Hive Metastore, Comparison with Traditional Databases, HiveQL, Tables, Querying Data and User Defined Functions) Hbase (H Basics, Concepts, Clients, Example, Hbase Versus RDBMS) Big SQL (Introduction), Interacting with Hadoop Ecosystem.	5	
6	Apache Spark: The basic usage of Spark and examples, how to use Spark and its different components.	6	
7	Data Analytics with R: Machine Learning (Introduction, Supervised Learning, Unsupervised Learning, Collaborative Filtering) Big Data Analytics with R, Integrating R and Hadoop.		
Course outcomes			
After completion of the course, a student would be able to:			
CO 1	Understand Big Data and its analytics in the real world		
CO 2	Analyze the Big Data framework like Hadoop and NOSQL to efficiently store and process Big Data to generate analytics		
CO 3	Design of Algorithms to solve Data Intensive Problems using Map Reduce Paradigm		
CO 4	Design and Implementation of Big Data Analytics using pig and spark to solve data intensive problems and to generate analytics		
CO 5	Implement Big Data Activities using Hive		
CO 6	Apply Machine Learning Techniques using R.		
Learning Resources:			
1.	V.K. Jain, Big Data and Hadoop, Khanna Publishing House, New Delhi(2017).		
2.	V.K. Jain, Data Analysis, Khanna Publishing House, New Delhi (2019).		
3.	Tom White “Hadoop: The Definitive Guide” Third Edit on, O’reily Media, 2012.		
4.	Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015.		
5.	Bahga, Arshdeep and Vijay Madiseti, Big data science & analytics: A hands-on approach,(1e),VPT,2016.		

6.	EMC Education Services, Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data,(1e),John Wiley & Sons, 2015.
7.	Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007.
8.	Jay Liebowitz, "Big Data and Business Analytics" Auerbach Publications, CRC press (2013)
9.	Tom Plunkett, Mark Hornick, "Using R to Unlock the Value of Big Data: Big Data Analytics with Oracle R Enterprise and Oracle R Connector for Hadoop", McGraw-Hill/Osborne Media (2013), Oracle press.
10.	Glen J. Myat, "Making Sense of Data", John Wiley & Sons, 2007
11.	Pete Warden, "Big Data Glossary", O'Reily, 2011.
12.	Michael Mineli, Michele Chambers, Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley Publications, 2013.
13.	Arvind Sathi, "Big Data Analytics: Disruptive Technologies for Changing the Game", MC Press, 2012

Name of the Course	Natural language processing
Course Code: PGCSPEC303B	Semester: 3rd
Duration: 6 months	Total Marks: 100
Teaching Scheme	Examination Scheme
Theory Contact Hrs.: 3 hrs./week	Mid Semester 1 Exam: 15 Marks
Tutorial Contact Hrs.: 0 hrs./week	Mid Semester 2 Exam: 15 Marks
Practical: 0 hrs./week	Other Assessment tools (Assignment, Quiz etc.): 20 Marks
Credit Points: 3	End Semester Exam:100 Marks (To be mapped into 50 marks)

Objective:	
1.	To get an overall idea of the characteristics of natural languages and the phases in NLP.
2.	To understand the issues in representation of texts in a computer, and the standards.
3.	To understand how words are formed in a natural language, and how to analyse words for NLP.
4.	To understand how the role of words or groups of words in a sentence have to be determined so as to recognise the syntactic structures of sentences.

Pre-Requisite	
1.	Machine Learning
2.	Python

Detailed Syllabus			
Module	Content	Hrs	Marks
1	Overview of Natural Language Processing: Text Analytics and NLP; Different NLP Steps; Tokenization; PoS Tagging; Stop Word Removal; Text Normalization; Spelling Correction; Lemmatization; Stemming; Named Entity Recognition (NER); Word Sense Disambiguation; Sentence Boundary Detection.	5	
2	Approaches for Feature Extraction: Data Categorization, Text Cleaning and Tokenization, extracting n-grams, Tokenizing Text, Regexp Stemmer, The Porter Stemmer, and Other Tokenizers Lemmatization, Singularization and Pluralization of Words, Language Translation, Removal of Stop Words from Text, Bag of Words (BoW), Creating a Bag of Words, Zipf's Law, Feature Extraction from Texts. Other Visualizations: Term Frequency-Inverse Document Frequency (TF-IDF), Feature Engineering, Word Clouds Dependency Parse Trees and Named Entities.	7	
3	Creating a Classifier for Text: Building a Text Classifier, Extracting Features, Engineering Features, Eliminating Correlated. Features, Eliminating Highly Correlated Features (Tokens), Finding the RMSE and MAPE of a Dataset, assessing a Model's Performance, and Carrying Out Dimensionality Reduction Using Principal Component Analysis.	5	
4	Topic Modelling: Topic exploration, converting unstructured to organized data, Algorithms for Topic-Modelling, The Operation of Latent Semantic Analysis (LSA) Using Latent Semantic Analysis to Examine Wikipedia World Cup Articles Latent Dirichlet Allocation (LDA): An Overview of Its Operation, Dirichlet Process and Distribution Using the LDA Model to Find Topics.	7	
5	Creation of Text, Summarization, and Vector representation: Vector, Document Vectors, Uses of Document Vectors, Converting News Headlines to Document Vectors, Finding Similar News Articles Using Document Vectors, Generating Text using Markov Chains, Text Summarization.	6	
6	Large Language Models (LLM), Concepts behind LLMs, Transformers: Transformer architecture, the encoder and the decoder, Computing the output of the self-attention layer, Embedding layers in the Transformer, Residuals and normalization. ChatGPT and Google BARD, GPT Models	6	

Course outcomes	
After completion of the course, a student would be able to:	
CO 1	Describe the concepts of Natural Language Processing.
CO 2	Demonstrate understanding of Approaches for Feature Extraction.
CO 3	Create a Classifier for Text
CO 4	Develop systems for Using APIs and Web Scraping to Gather Text Data.
CO 5	Describe Topic Modeling
Learning Resources:	
1.	Daniel Jurafsky and James H Martin. Speech and Language Processing, 2e, Pearson Education, 2009
2.	Siddiqui T., Tiwary U. S.. Natural language processing and Information retrieval, OUP, 2008

Name of the Course	Internet of Things
Course Code: PGCSPEC303C	Semester: 3rd
Duration: 6 months	Total Marks: 100
Teaching Scheme	Examination Scheme
Theory Contact Hrs.: 3 hrs./week	Mid Semester 1 Exam: 15 Marks
Tutorial Contact Hrs.: 0 hrs./week	Mid Semester 2 Exam: 15 Marks
Practical: 0 hrs./week	Other Assessment tools (Assignment, Quiz etc.): 20 Marks
Credit Points: 3	End Semester Exam:100 Marks (To be mapped into 50 marks)

Objective:	
1.	To understand the application areas of IOT .
2.	To realize the revolution of Internet in Mobile Devices, Cloud & Sensor Networks.
3.	To understand building blocks of Internet of Things and characteristics.

Pre-Requisite	
1.	Computer Networks

Detailed Syllabus			
Module	Content	Hrs	Marks
1	Introduction: What is IoT, Genesis of IoT,Convergence of IT and oT, IoT Challenges, IoT Architecture and Design, Physical & Logical design of IoT, A simplified IoT Architecture.	9	
2	Major components of IoT: IoT enabling Technologies Sources of IoT, M2M Communication, M2M Architecture, Difference between M2M and IoT, Data and Analytics for IoT, An Introduction to Data Analytics for IoT, Machine Learning, Big Data Analytics Tools and Technology.	8	
3	Smart Objects: The “Things” in IoT: Sensors, Actuators, and Smart Objects, Sensor Networks, Connecting Smart objects, Working Principles of sensors, Introduction to Arduino Uno, and Raspberry PI, Features and application of Arduino uno.	6	
4	Securing Internet of Things: Security Requirements in IoT Architecture - Security Concerns in IoT Applications. Cryptographic primitives and its role in Digital Signatures and light weight cryptography, Cryptographic controls built into IoT messaging and communication protocols.	8	
5	Recent trends in smart sensor for day-to-day life: Real world applications for IoT: IoT based home automation with security features, Smart farming: IoT based system for smart agriculture, IoT application to improvise industrial automation, – Smart Healthcare systems etc.	5	

Course outcomes	
After completion of the course, a student would be able to:	
CO 1	Explain general concepts of Internet of Things (IoT).
CO 2	Construct various M2M and IoT architectures.
CO 3	Analyze the deployment of smart objects and the technologies to connect them to network.
CO 4	Evaluate security principles and methodologies using Cryptography for Internet of Things.
CO 5	Design different real-world applications.

Learning Resources:	
1.	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry,"IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things”.
2.	Srinivasa K G, “Internet of Things”,CENGAGE Leaning India, 2017 .
3.	Yasuura, H., Kyung, C.-M., Liu, Y., Lin, Y.-L., Smart Sensors at the IoT Frontier, Springer International Publishing.
4.	Jeeva Jose, Internet of Things, Khanna Publishing House

5.	Vijay Madiseti and Arshdeep Bahga, “Internet of Things (A Hands-on-Approach)”, 1 st Edition, VPT, 2014. (ISBN: 978-8173719547)
6.	Raj Kamal, “Internet of Things: Architecture and Design Principles”, 1st Edition, McGraw Hill

Name of the Course	Distributed Database
Course Code: PGCSPEC303D	Semester: 3rd
Duration: 6 months	Total Marks: 100
Teaching Scheme	Examination Scheme
Theory Contact Hrs.: 3 hrs./week	Mid Semester 1 Exam: 15 Marks
Tutorial Contact Hrs.: 0 hrs./week	Mid Semester 2 Exam: 15 Marks
Practical: 0hrs/week	Other Assessment tools (Assignment, Quiz etc.): 20 Marks
Credit Points: 3	End Semester Exam: 100 Marks (To be mapped into 50 marks)

Objective:	
1.	To explain the basic principles of distributed database systems and its architectures
2.	To evaluate the efficiency of distributed database design and the distributed query processing plans
3.	To handle different concurrency control anomalies and maintain the reliability of distributed transactions.
4.	To discuss the overall concept of parallel database systems

Pre-Requisite	
1.	Database Management System
2.	Computer Network

Detailed Syllabus			
Module	Content	Hrs	Marks
1	Introduction to Distributed Database Management Systems (DDBMS): Basic idea of DDBMS, homogenous, heterogeneous and federated DDBMS, DDBMS fragmentation, replication and allocation concept, global and local schema, key advantages and reliability of DDBMS, different layers of distribution transparency, complications of DDBMS, reference architecture for distributed DBMS- client/server, peer-to-peer, and multidata base systems, global directory issues in distributed DBMS.	6	
2	DDBMS Design Aspect: Design issues of DDBMS, non-replicated, non-fragmented, fully replicated, partially replicated DDBMS, top down & bottom up approach of DDBMS design, reasons for data fragmentation, degree of fragmentation, different type of fragmentations- primary and derived horizontal fragmentation, vertical fragmentation mixed or hybrid fragmentation, correctness rules of fragmentation, data fragment allocation and its associated issues, basic idea of view management- views in centralized and distributed database.	6	
3	DDBMS Query Processing: Query processing issues in DDBMS, objectives of distributed query processing, different layers of distributed query processing- query decomposition, data localization, global query optimization, distributed query execution, centralized query optimization- main objectives, query processing with translation and optimization steps, concept of distributed query optimization- distributed query processing architecture, mapping global queries into local queries, main issues of distributed query optimization.	6	
4	Distributed Transaction Management & Reliability Aspects: Concept of transaction & its properties, serial and parallel schedule, locking based concurrency control- 2PL & strict 2PL, distributed concurrency control algorithms- centralized & distributed 2PL, time stamp-based concurrency control algorithms, deadlock detection & resolution- centralized, hierarchical and distributed deadlock detection, concept of reliability & different types of failures in DDBMS, idea of local recovery manager, distributed reliability protocols- centralized & distributed 2PC, termination protocol for 2PC-coordinator & participant timeout, non-blocking commit protocol, voting protocol.	12	
5	Parallel Database Systems: Basic idea, objectives, parallel database system architectures- functional architecture, shared memory, shared disk, shared nothing & hybrid architecture, concept of database cluster & architecture, parallel data placement- round-robin, hash and range based partitioning, parallel query processing- intra & inter operator parallelism.	6	

Course outcomes	
After completion of the course, a student would be able to:	
CO 1	Explain the overall concept of distributed database systems and its different associated components
CO 2	Choose suitable design strategies for the distributed database systems in the aspects of data storage and views
CO 3	Analyze different query processing plans applicable for the distributed database systems
CO 4	Compare various concurrency control mechanisms and recovery issues for distributed database systems

CO 5	Discuss the main idea and features of parallel database system in the aspect of performance benefits
Learning Resources:	
1.	Stefano Ceri, Guiseppe Pelagatti, “Distributed Databases: Principles and Systems”, McGraw Hill Education, Indian Edition, 2017.
2.	M. Tamer Ozsu, Patrick Valduriez, “Principles of Distributed Database Systems”, Springer, Third edition, 2011.
3.	Saeed K. Rahimi, Frank S. Haug, “Distributed Database Management Systems: A Practical Approach”, Wiley-IEEE Computer Society, Aug, 2010, Print ISBN:9780470407455, Online ISBN:9780470602379
4.	Chhanda Ray, “Distributed Database Systems”, Pearson Education India, 1st Edition, Kindle Edition, 2009, ISBN- 9788131727188, 8131727181.
5.	Sachin Deshpande, “Distributed Databases”, Dream tech Press, Kindle Edition, 2014, ISBN 13: 9789351197201

4Th SEM							
SL. NO.	PAPER CODE	PAPER NAME	L	T	P	CONTACT HRS./WEEK	CREDIT
SESSIONAL / PRACTICAL							
01	PGCSDIS402	Dissertation-II	0	0	30	30	15
02	PGCSCVV401	Comprehensive Viva-Voce	0	0	0	0	2
		Total	0	0	30	30	17